

If pictures are needed/relevant, please provide them with your solutions.

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- Recall: $V_B - V_A = - \int \mathbf{E} \cdot d\mathbf{l}$, with the line integral proceeding along any curve starting at point A and ending at point B .

2. The electric field in some region is $\mathbf{E} = E_0\hat{i} + 2E_0\hat{j}$.

- (a) [**3 pts.**] A planar surface of area 2α lies in the x - y plane, i.e., the normal to the surface is in the z direction. Find the magnitude of the flux of the electric field through this surface.
- (b) [**5 pts.**] A planar surface of area 6α lies parallel to the y - z plane. Find the magnitude of the flux of the electric field through this surface.

3. Recall that the current through a surface (e.g., through a cross-section of a current-carrying wire) is the surface integral of the current density.
- (a) **[6 pts.]** A current I is uniformly distributed over the cross-section of a wire. The cross-section is circular, with radius R . Find the current density J , as a function of I and R .
- (b) **[6 pts.]** Now consider instead the situation that the current density is larger near the outer region of the wire cross-section than at the center: $J = kr$, where r is the radial distance from the center of the wire. Find the current I , as a function of k and R . You will probably need to do a two-dimensional integral in polar coordinates. (In two-dimensional polar coordinates, the area element is $dS = r dr d\theta$.)
4. A copper conductor has 8.5×10^{28} electrons per cubic meter available to carry current, i.e., the carrier density is $8.5 \times 10^{28} \text{m}^{-3}$. The drift velocity is found to be $3 \times 10^{-6} \text{m/s}$.
- (a) **[2 pts.]** Look up and report the charge of an electron in SI units.
- (b) **[4 pts.]** Calculate the current density J .
5. **[7 pts.]** The current density in some region is found to be

$$\mathbf{J} = \frac{K_0 y}{\sqrt{x^2 + y^2}} \hat{i} - \frac{K_0 x}{\sqrt{x^2 + y^2}} \hat{j}$$

Use the continuity equation to show that the charge density in this region does not change with time.